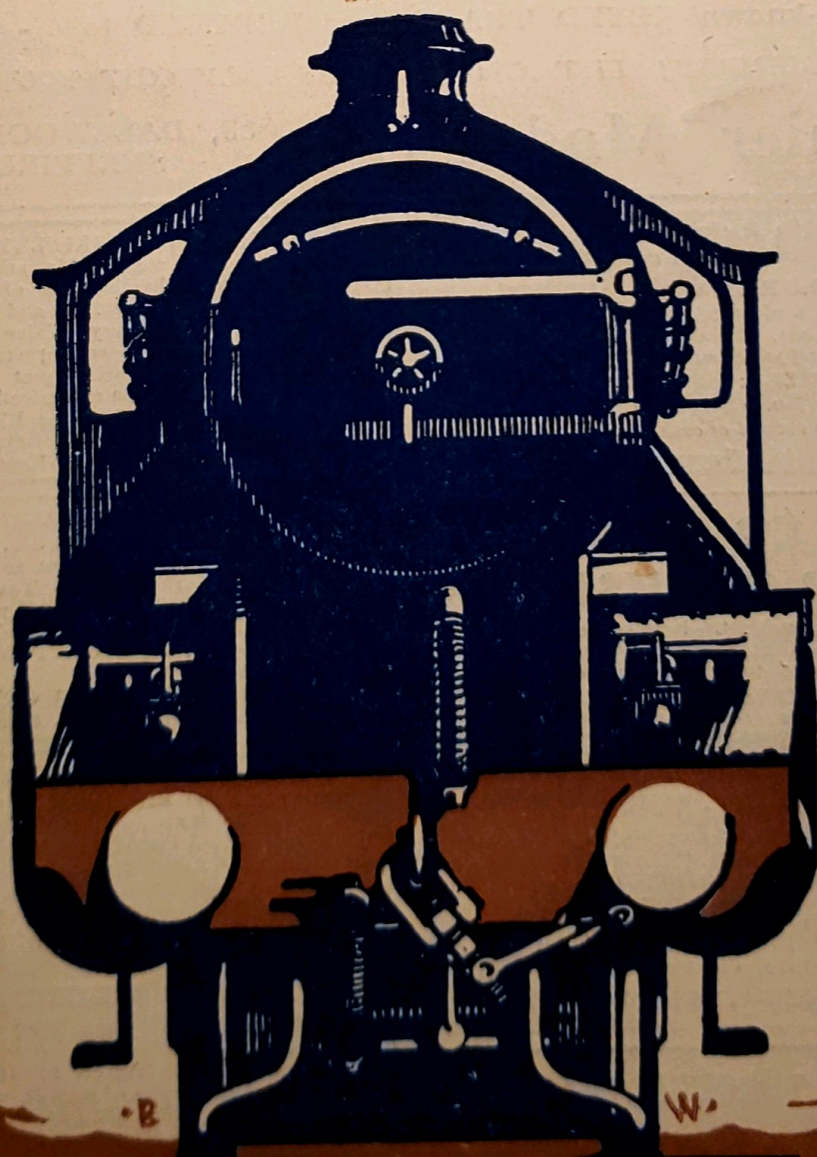


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A Rebuilt 2 in. Gauge Garden Railway

Describing some Free-lance designs of Bridges and Viaducts for the climatic conditions of an outdoor line

By GORDON HEMM, D.Arch., A.R.I.B.A., A.M.T.P.I.

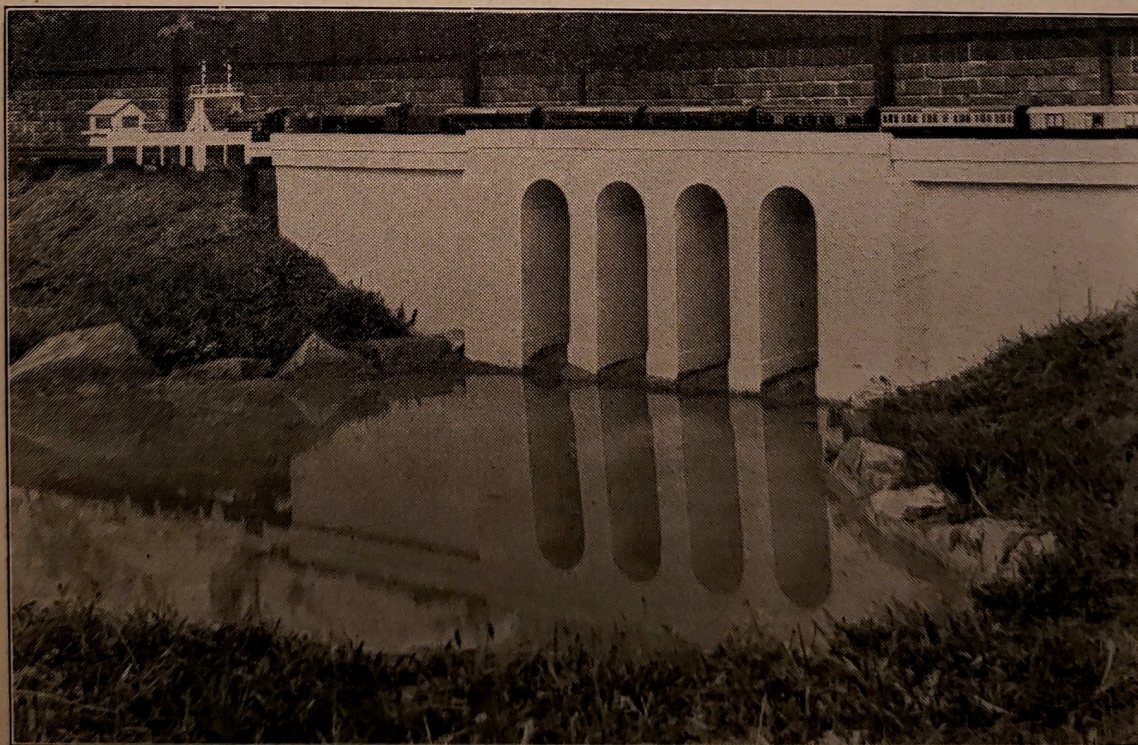
IT is some five years since I began to think seriously of building some kind of model railway. As a boy I had a passion for everything connected with the iron road, but during the intervening years, the necessity for long study in a profession that has something in common with railways, time for model-making was out of the question. That period commenced the breaking away of that boyish desire for the possession of some fine miniature railway, and a lapse of some twenty years went by before that latent craving for railwayism was revived.

The first important question to be settled was the kind of railway to develop, and the opportunity of the purchase of a second-hand "Atlantic" locomotive in 2" gauge seemed to

settle this difficult point. The lack of space indoors also influenced the building of an outdoor line, the results of which were illustrated in the *MODEL RAILWAY NEWS* for April, 1930.

After two years' running it was found necessary to reconstruct the line on an entirely new site, and the transfer meant removing the whole equipment a distance of some ten miles away. This was no small undertaking, as the concrete viaduct and rustic brick tunnel entailed much labour in the transit to the new garden, besides the abundance of timber which was used as supports for the actual track.

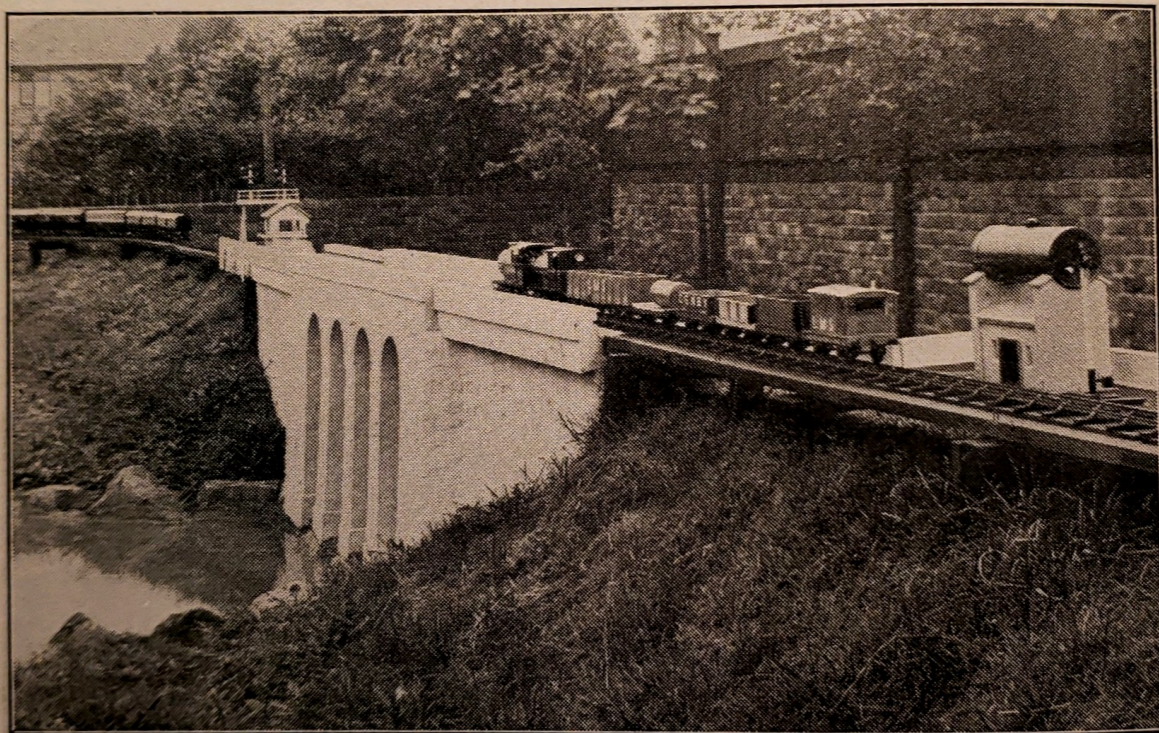
One difficult consideration in planning the new layout, was the question as to the height that the track should be elevated above ground level. For realism, undoubtedly a ground level



The Thames-Forth express passing over the concrete viaduct. Note the reflection in the pool

development has advantages, for with the raised position, the supports naturally take away somewhat from the effect of the real thing. The matter was settled by the fact that the original concrete viaduct, which in its original position had been made some 3' high, owing to the natural slope of the ground, was re-erected on ground level, so this formed the datum level for the general level of the track. One point in favour of the raised track is that any undue stooping is overcome, a physical position which is very tiring.

composed of arches, with pylon terminals, and spanned in the central by a metal lattice bridge, which combines well with general composition. This is over 13' in length, and 2' in width, and the total weight is several hundred-weight. It is to be hoped, at some future date, to give scenic interest by constructing a miniature valley in the centre of the group, and thus provide a logical reason for the planning of this feature. The facing of the concrete is treated with a very fine preparation called "Imstone," made by the Walpamur Company, which gives a



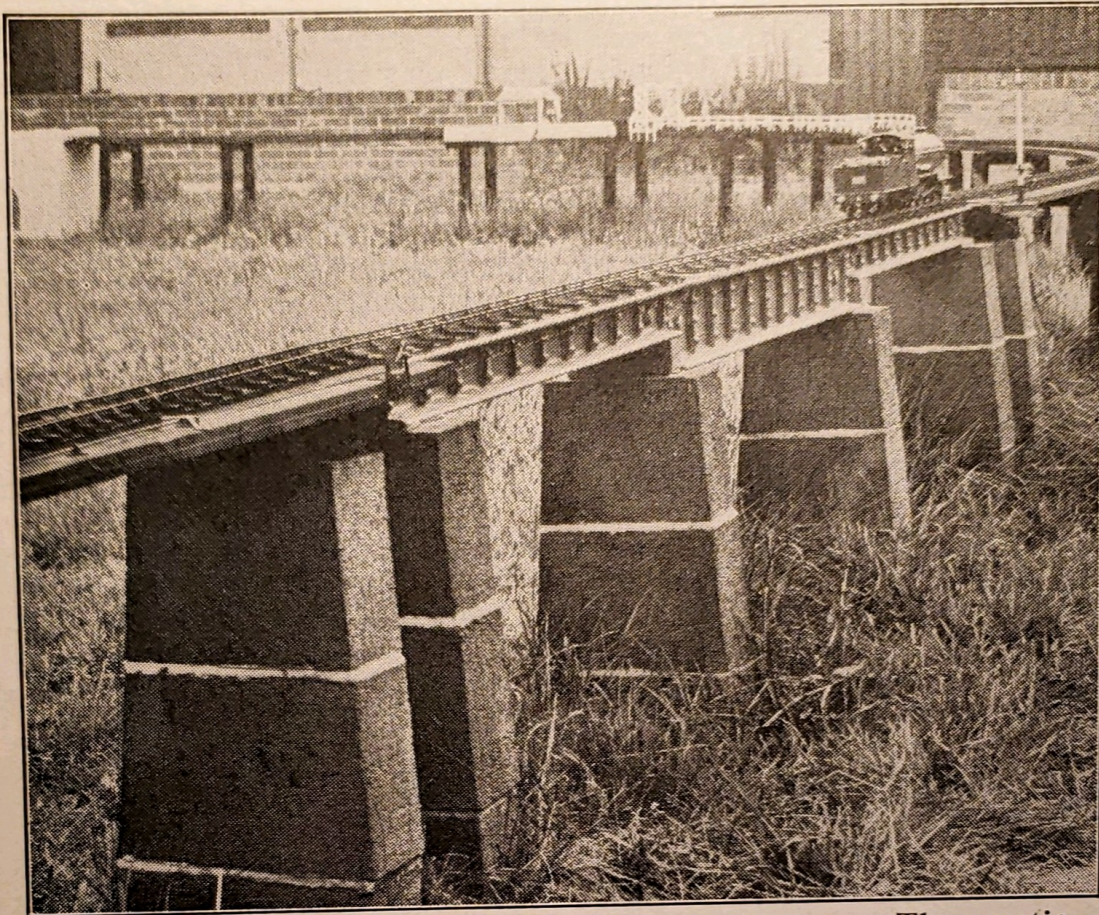
A goods train approaching the viaduct

The length of line has been more than doubled, and some 300' of track forming a continuous run can now be undertaken by large and small locomotives. This is a great advantage over the old layout, as owing to the additional area of garden, wider sweeps of curves have been made possible.

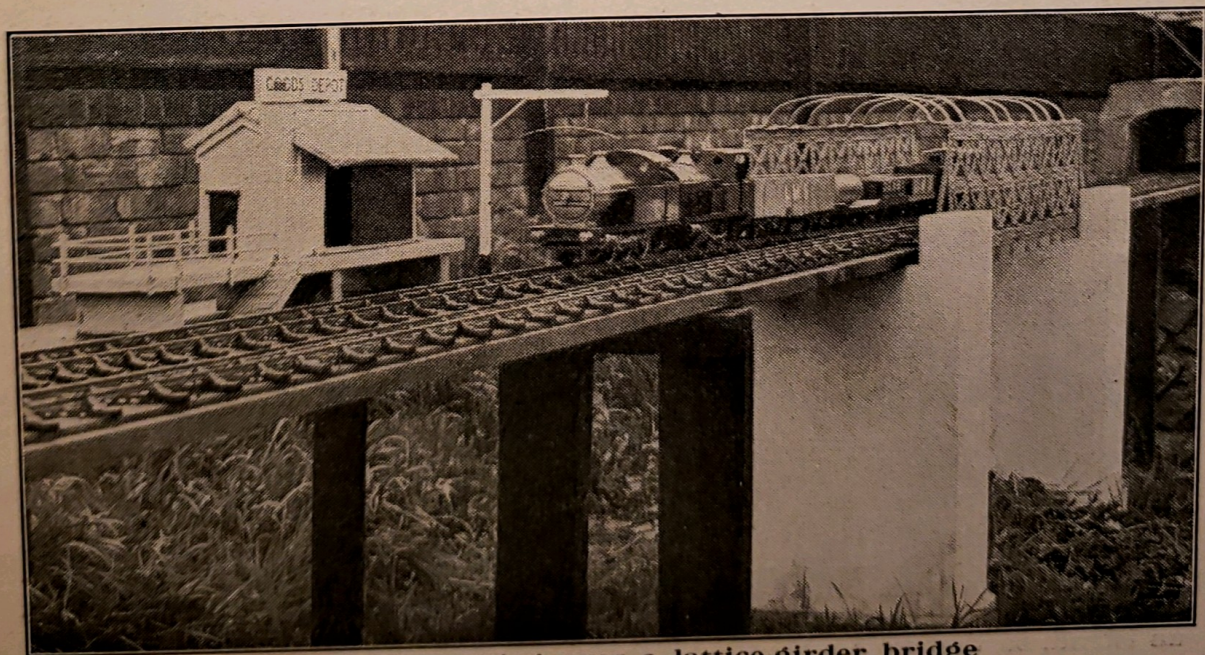
Greater garden space has given scope for more spectacular features, with the result that bridge building has been given due consideration. There are two bridges, besides the viaduct mentioned above; one, a concrete structure

remarkable likeness to stone, besides having the hard properties of good weathering stone.

The second bridge is conceived on different lines, and I must confess that American precedent gave me the inspiration in this case. Those fine plate girder bridges, which feature so many of those romantic valleys of the States, infused me with enthusiasm, but unfortunately I had no romantic valley to give that picturesque setting, and all I could do was to build up my efforts on level ground. A certain



A model bridge based on American prototypes. The rustic piers were specially cast and form logical supports for the weight they have to withstand



A goods train passing over a lattice girder bridge

realism to the prototype is the suggestion of built-up plates of steel, formed by roll steel joists, with vertical strips applied to the outer web. The supports were specially ordered, and they have been detailed to taper upwards, and so in model form they imitate the practice which all true engineers incorporate in the real pier structures.

Perhaps the feature of the railway that makes its greatest appeal to most visitors, is the concrete viaduct, which is enhanced in effect by the building of a small pool to the fore, and thus interesting reflections are to be seen

possible to build up realistic approaches to the station, and so the type with two platforms, making a complete double road station, would add considerably to the layout. But when one has to deal with work on a larger scale, the island scheme has many merits. The station in question is by no means small, for it exceeds 14' in length, and for weathering, has been built in three sections. It contains the necessary accommodation required for a building of this kind, and the spectacular accessories, such as telephone boxes, weighing machines and chocolate machines are much in



A concrete and lattice girder bridge, 13 ft. long, the latest addition to the line

at different times of the day. I strongly advise anyone interested in the erecting of an outdoor line to experiment with concrete work; they will find it will have untold possibilities for very striking effects of realism.

Stations at present are limited to one, for the simple reason that it takes a considerable time to construct this type of building to the scale of the 2" gauge. In designing the station to a line that is raised on supports, the most effective type is the one forming an island, so that no backing is to be seen. On a smaller scale, it would be

evidence. There are also posters, which for protection against the elements are glazed. The roof is covered with lead, while the superstructure of the station is wood, stoutly detailed, and covered with bright colours of paint.

One must remember that in the design of such items as bridges, stations, signal-boxes, etc., on a scale of this size, it well repays to carry out the work on the free-lance principle. The fact that one is always contending with weather conditions, makes it impossible to design railway buildings to the exact prototype of

the real thing. Therefore, in my stations, buildings, and other forms, a free-lance spirit pervades them all. Another point of merit is the fact that a greater scope for originality can be displayed when one is not down to a miniature reduction from actual railway buildings.

The design of the signal boxes echo something of the standard box, yet designed to a consistent scale, they denote freedom of concept. Likewise, the goods depot and the oil

storage building, these are characteristic efforts of such features of the line. The signals work from the lever equipment in each box, and while the points function from the same centres, the actual operation of the block system was found to be impossible.

The rolling stock is varied, consisting of G.W.R. and L.M.S. vehicles. The locomotives are Messrs. Bassett-Lowke's, of Northampton, and are all constructed on the high pressure system of working.

Real Railway Topics

By J. N. MASKELYNE, A.I.Loco.E.

Another New L.M.S. Design

For the L.M.S. Northern Counties Committee, Ireland, Mr. W. A. Stanier has designed and built, at the Derby Works, four very handsome 2-6-0 type tender engines to suit, of course, the Irish standard rail gauge of 5' 3". The first engine and tender of this new class were loaded on four-wheeled-bogie wagons for conveyance from Derby to Heysham, where they were transferred to the boat for shipment to Belfast. On arrival at Belfast, the engine and tender were unloaded and wheeled on the quayside, and then taken on their own wheels to the Northern Counties Committee Shops at Belfast. The photograph and diagram, herewith, illustrate the general appearance of the new design.

General Features

The boiler is a slight modification of a standard type with Belpaire firebox; advantage has been taken, how-

ever, of the wide distance between the frames (due to the 5' 3" gauge) to widen the firebox and grate, with the result that a grate area of 25 sq. ft. has been made possible with a firebox length of only 7' 6" outside. Top feed apparatus is provided, but, in this case, the clack valves are housed in a separate casing mounted on the top of the boiler barrel and situated between the dome and the back of the smokebox. The superheater and other boiler fittings are, as far as possible, of standard L.M.S. types. Two simple cylinders, 19" diameter by 26" stroke, are provided, driving on to the middle coupled axle, and Walschaerts valve-gear operates the 9" diameter piston valves, of which the travel, in full gear, is 6 $\frac{3}{8}$ ". The very commodious cab has two sliding windows on each side, while the general arrangement of the footplate fittings, which includes tip-up seats, has been very carefully planned to suit the convenience of

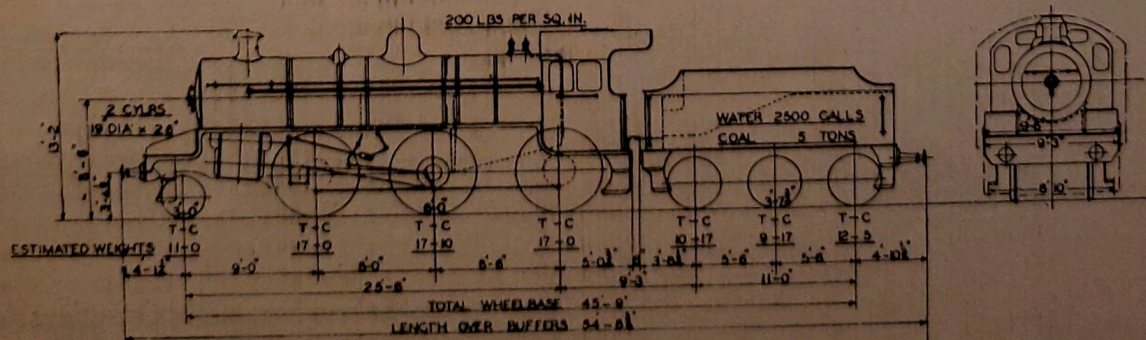


Diagram of new 2-6-0 locomotive for the L.M.S.R., N.C.C., Ireland.